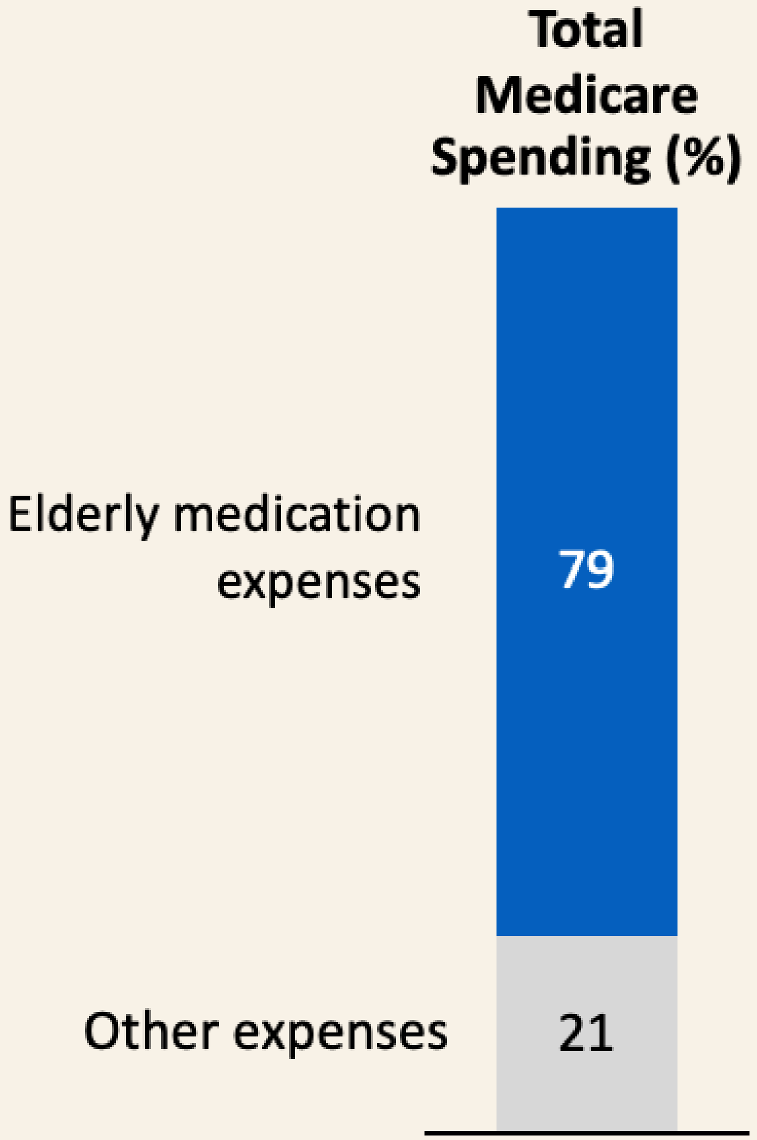
Medication Management for Older Adults (iMAP) trial Projects

Group 4_Final Presentation

Improving elderly’s medication use is a major and growing public health challenge in U.S.



Economic burdens

Elderly’s complex medication regimens contributes to large proportion of medicare spending

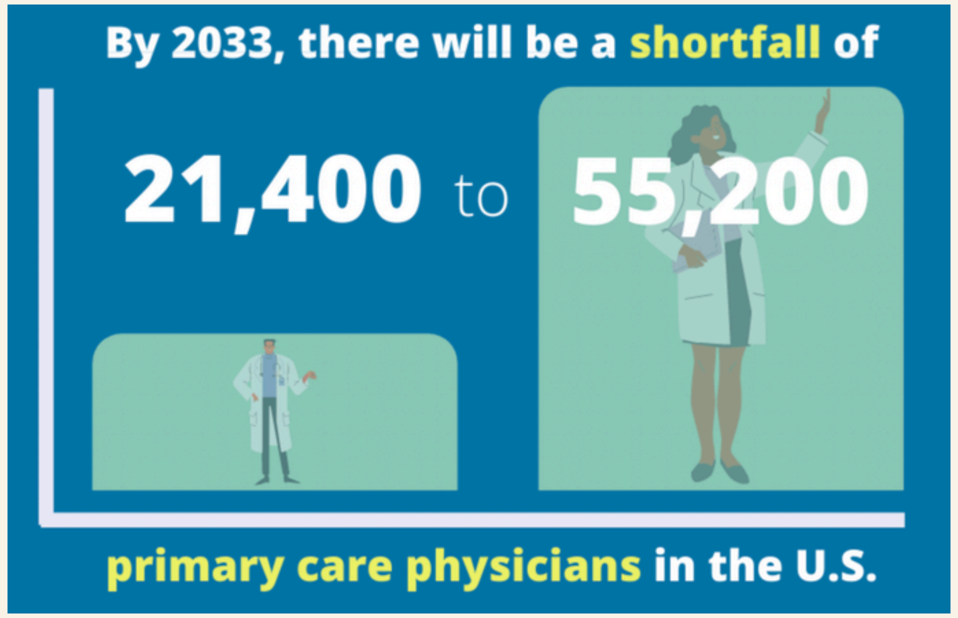
Elderly’s Common Medical Related Problem (MRPs)

1. Underuse of medications
2. Overuse of medications
3. Misuse of medications

Public health burdens

Despite tremendous financial investments, their medication usages are often suboptimal, negatively affecting health and well-being

Supply and Demand for Primary Care Physicians



Healthcare burdens

Primary care physicians are overloaded, result in deprioritized elderly’s comprehensive medication therapy management (MTM)

MTM: Medication Therapy Management | **MRP:** Medication-Related Problems

Existing Medication Therapy Management (MTM) targets select MRPs and limits in scope, iMAP possess a comprehensive and scalable solution

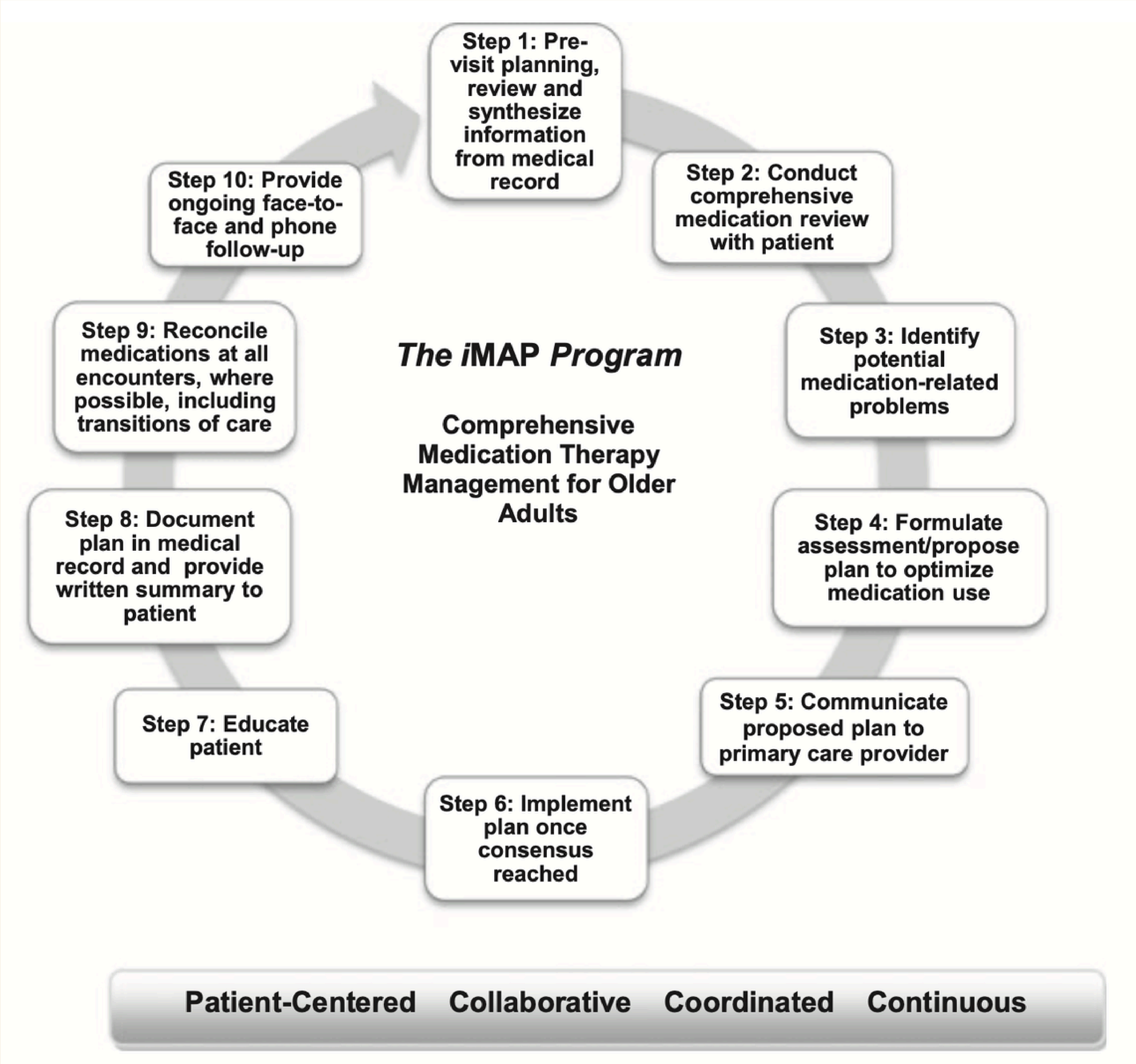
	Description	Limitation
Existing MTM: research	**Targets single MRP**, such as inappropriate prescribing, to improve medication use	**Unlikely to optimize overall medication use** in individuals, as older adults often have multiple chronic conditions
Existing MTM: national program	Federal government offers annual comprehensive medication reviews, conducted by qualified health care professionals, to **identify / resolve MRPs**	They are **limited in scope** (reaching <15% medicare beneficiaries) **and intensity** (typically offered remotely and independent of primary care practice)

→

Solution
iMAP – a comprehensive MTM with scalability potential • Involves clinical pharmacists working as part of the primary care team to provide direct patient care • Optimizes medication use • Maximizes clinical outcomes

MTM: Medication Therapy Management | **MRP:** Medication-Related Problems | **iMAP:** Individualized Medication Assessment and Planning

iMAP program: A 10-step Procedure to Improve Medication Use



A **reliable and standardized approach** for clinical pharmacists to classify MRPs and associated recommendations

Next Step on iMAP Intervention

2013 iMAP pilot study found that iMAP is **feasible** and showed promise as an **effective** intervention program, providing motivation for an additional trial that:

- Has a **randomized controlled** design
- Is conducted across **multiple sites with more patients**
- Includes **patient self-reports** of acute health utilization outside the primary point-of-capture

Research Question

iMAP intervention is **superior** to SC in two community-based primary care practices in leading to better health outcomes

02 - Study Design/Methods

- **Study Population**
  - Adults ≥65 years
  - ≥3 medications (≥1 prescription)
  - Life expectancy ≥12 months
  - Receiving ongoing primary care
  - Comparator: iMAP intervention vs Standard Care
- **Sample Size:** 3504 based on our calculation

- **Primary Outcome**
  - **Aim 1:** Determine whether the iMAP intervention reduces the total number of medication-related problems (MRPs) compared with standard care in 6 months.
  - **Aim 2:** Determine whether the iMAP intervention reduces the risk of first acute health services compared with standard care within 12 months.

Intention-to-Treat Analysis

- All participants analyzed per original randomized assignment
- Unit of analysis: individual patient
- Two-sided tests at 5% significance level
- 95% confidence intervals reported for all estimates
- Model assumptions checked; alternatives used if violated

Baseline Comparability

- Age, biological sex, race
- Total medications and prescriptions
- Healthcare utilizations in the prior year
- Number of comorbid conditions

Categorical variables: frequencies and percentages
Continuous variables: mean, standard deviation, range
Visual comparison via boxplots

Primary and Secondary Analysis Methods & Assumption Checks

Outcome	Explanation	Method	Key Model Features	Treatment Effect Measures
Aim 1/2: MRP Count	Total MRPs at 6/12 months	Generalized Linear Mixed Model	Treatment-by-time interaction; patient-level random intercept; adjusted for site and race	Mean difference with 95% confidence interval
Aim 3: Time to First Utilization	Time to first service utilization within 12 months	Cox Proportional Hazards Regression	Treatment group plus baseline covariates; adjusted for site and race	Hazard ratio with 95% confidence interval
Aim 4: Utilization Count	Total service utilization within 12 months	Negative Binomial	Treatment group, stratification variables, and log of follow-up time in months	Incidence rate ratio with 95% confidence interval

GLM Assumptions

- **Normality:** Check residual distribution through Q-Q plots and Shapiro-Wilk test; if violated, consider transformations or non-parametric methods
- **Model fit:** Residual diagnostics to assess linearity and homoscedasticity

Cox Regression Assumptions

- **Proportional hazards:** Schoenfeld residual test and visual inspection; if violated, use restricted mean survival time
- **Censoring rules:** Death: censored at time of death; no event by 12 months: censored at end of follow-up
- ***Competing risks:*** *As a sensitivity analysis, treat death as a competing event rather than censoring*

Negative Binomial

- **Overdispersion Check:** Variance-to-mean ratio and likelihood ratio test comparing NB vs. Poisson

Missing Data Handling

Primary Analysis
Assume data are missing completely at random (**MCAR**); analyze available cases without imputation

Assess the Assumption
Compare baseline characteristics between participants with complete vs. incomplete data
Report the percentage of missing data by treatment arm

< 5% missing

Negligible
No additional action needed; primary analysis results are reliable

5 - 10% missing

Moderate
Adjust for baseline covariates associated with missingness in the analysis model

>10% missing

Substantial
Conduct multiple imputation; results combined using Rubin's rules

Note: If missingness is found to be associated with baseline factors at any level, the MCAR assumption may not hold. In that case, adjust for related covariates or conduct multiple imputation regardless of the missing percentage.

RCT is expected to enroll at least 3,504 patients

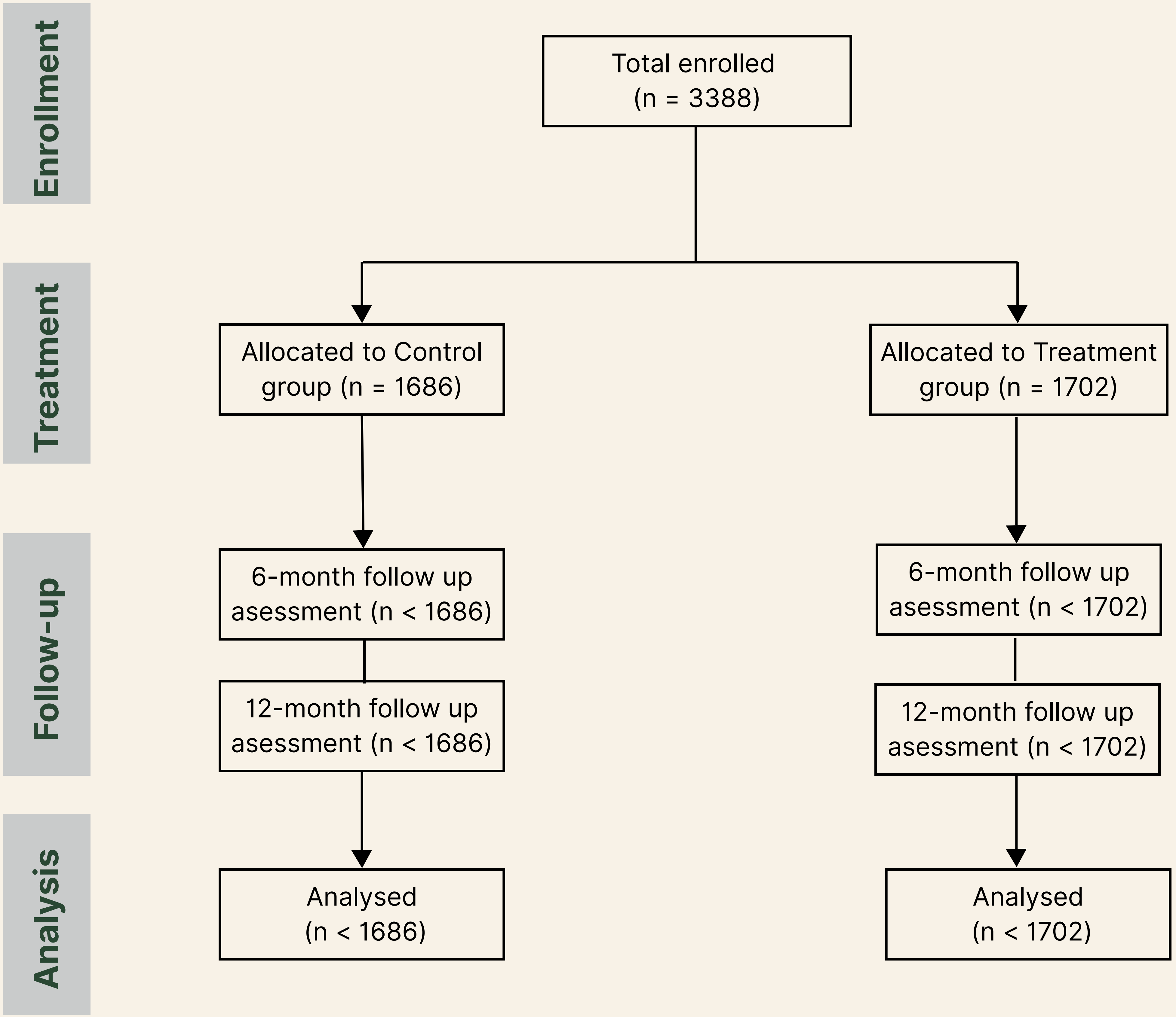
- **Guiding Assumptions**
  - Assume smaller effect size and larger variance than pilot study (conservative planning)
  - Assume type I error of 5% (two-sided) and power of 90% for each aim

- **Variables Needed for Power Analysis**

Variables		Pilot Study	RCT estimate	Sample Size
Number of MRPs (6 months)	Effect Size	3.2	1	310
	Effect's Standard Deviation	1.5	2.5	
1st Service Utilization (12 months)	Hazard Ratio	65%	80%	3504
	Event Rate		28.4%	
Attrition Rate		14.1%	15%	

Note: arrows represent upgrade/downgrade in a manner that gives conservative estimates

Figure 1: Participant Flow



Participant Characteristics by Treatment Group

Variable	Level	Overall (N=3388)	iMAP (N=1702)	Standard Care (N=1686)
Race (%)	non-White	712 (21.0)	359 (21.1)	353 (20.9)
	White	2676 (79.0)	1343 (78.9)	1333 (79.1)
Sex (%)	Female	2195 (64.8)	1094 (64.3)	1101 (65.3)
	Male	1193 (35.2)	608 (35.7)	585 (34.7)
Site (%)	Site A	2202 (65.0)	1103 (64.8)	1099 (65.2)
	Site B	1186 (35.0)	599 (35.2)	587 (34.8)
Age (mean, SD)	—	78.55 (7.91)	78.51 (7.95)	78.59 (7.86)
Total medicines (mean, SD)	—	10.43 (3.14)	10.48 (3.12)	10.37 (3.16)
Prescribed medicines (mean, SD)	—	6.92 (2.47)	6.98 (2.44)	6.87 (2.50)
Prior health utilizations (mean, SD)	—	1.64 (2.02)	1.62 (2.00)	1.67 (2.04)
Comorbid conditions (mean, SD)	—	4.83 (1.30)	4.86 (1.30)	4.81 (1.30)

SD = Standard Deviation

Participant Characteristics by Outcome Missingness

Variable	Level	Observed (N=3238)	Missing at 6 month (N=150)
Treatment (%)	iMAP	1625 (50.2)	77 (51.3)
	Standard Care	1613 (49.8)	73 (48.7)
Race (%)	non-White	677 (20.9)	35 (23.3)
	White	2561 (79.1)	115 (76.7)
Sex (%)	Female	2107 (65.1)	88 (58.7)
	Male	1131 (34.9)	62 (41.3)
Site (%)	Site A	2115 (65.3)	87 (58.0)
	Site B	1123 (34.7)	63 (42.0)
Age (mean, SD)	—	78.32 (7.86)	83.41 (7.35)
Total medicines (mean, SD)	—	10.36 (3.12)	11.77 (3.22)
Prescribed medicines (mean, SD)	—	6.90 (2.46)	7.54 (2.79)
Prior health utilizations (mean, SD)	—	1.65 (2.03)	1.51 (1.79)
Comorbid conditions (mean, SD)	—	4.80 (1.30)	5.53 (1.20)

SD = Standard Deviation

Result: Effect on MRPs at 6 and 12 Months

MRP Count Analysis

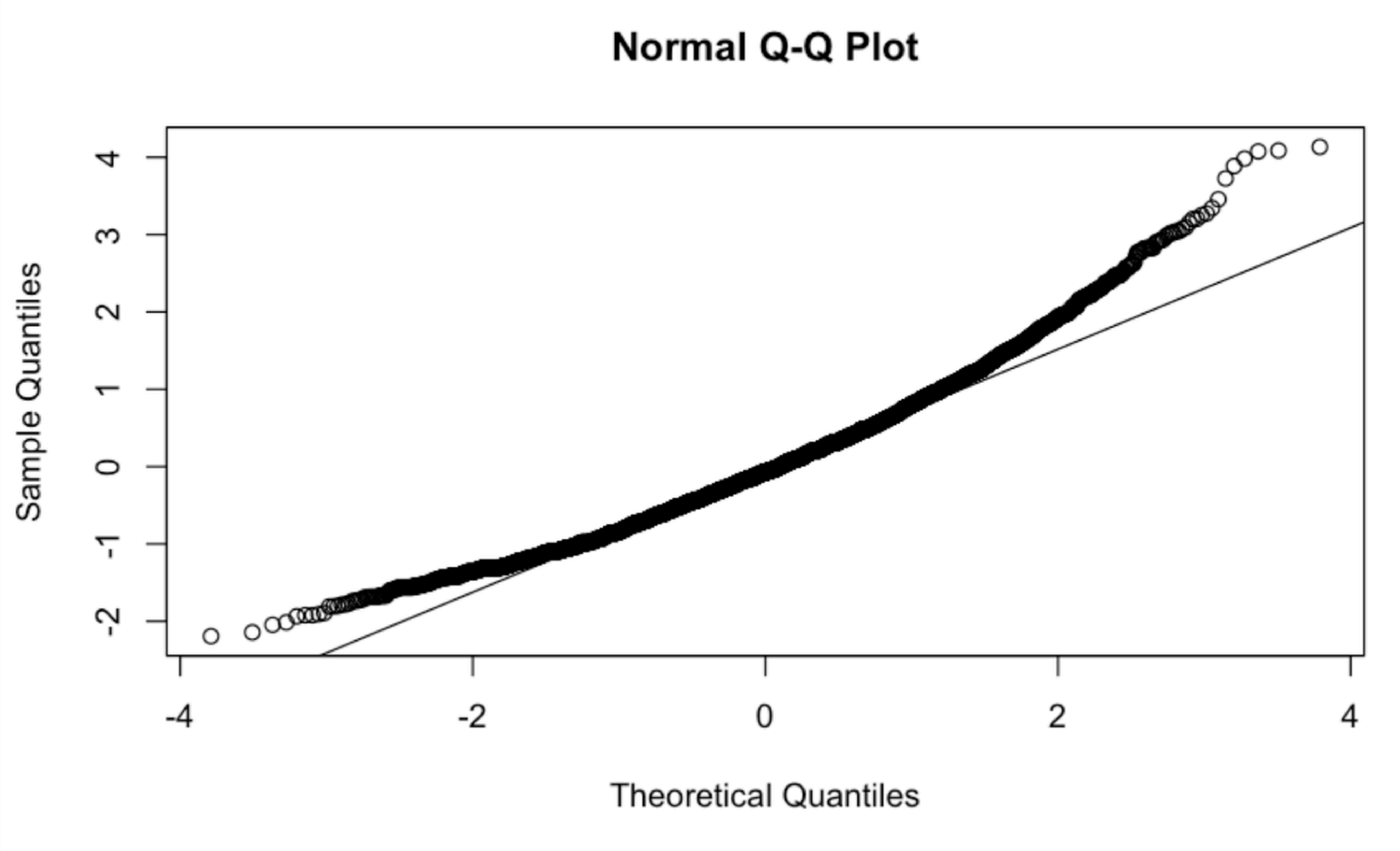
Time Point	Comparison (iMAP - SC)	Estimate	SE	95% CI	p-value
6 months	iMAP - Standard Care	-1.15	0.117	(-0.92, -1.38)	<0.0001
12 months	iMAP - Standard Care	-1.71	0.13	(-1.45, -1.97)	<0.0001

Sensitivity Analysis

Time Point	Comparison (iMAP - SC)	Estimate	SE	95% CI	p-value
6 months	iMAP - Standard Care	-1.16	0.115	(-0.94, -1.39)	<0.0001
12 months	iMAP - Standard Care	-1.75	0.135	(-1.48, -2.01)	<0.0001

R software used

Assumption Check - Normal Distribution

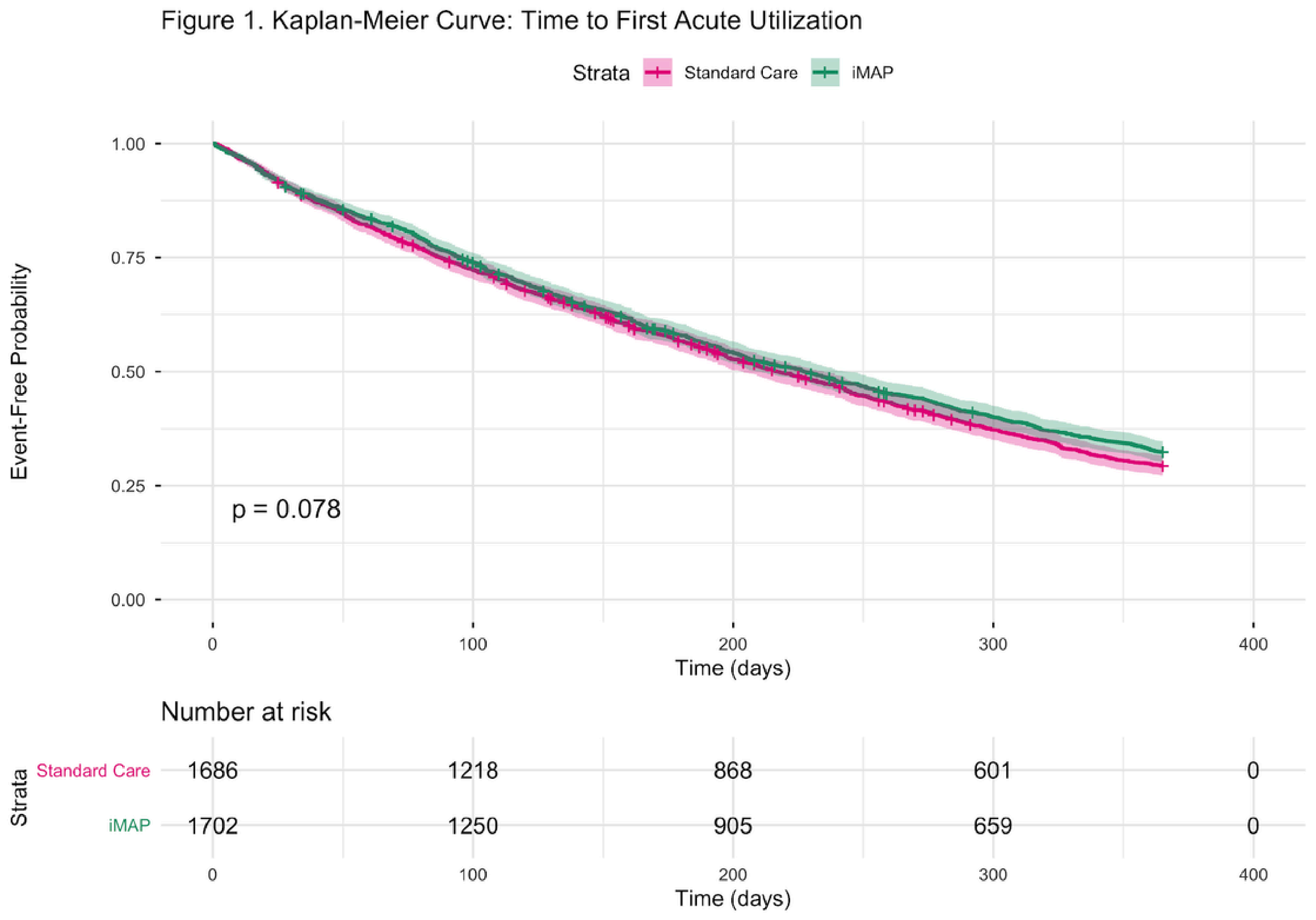


R software used

Result: Effect on Risk of Service Utilization

Cox PH Model (Time to First Acute Health Service Utilization)

Covariate	Hazard Ratio (HR)	95% Confidence Interval	p-value
iMAP vs. Standard Care	**0.929**	(0.856, 1.008)	**0.076**



iMAP did not significantly delay the time to first acute care event (Log-rank p = 0.078)

R software used

Result: Effect on Number of Service Utilization

	Standard Care (N = 1,686)	iMAP (N = 1,702)
Mean utilizations per person (SD)	1.17 (1.17)	1.07 (1.11)
Median utilizations per person	1	1
Total utilizations across arm	1,973	1,825
Mean follow-up time (months)	11.8	11.8

Negative Binomial Model - (Count of Acute Health Service Utilization)

Covariate	Incidence Rate Ratio (IRR)	95% Confidence Interval	p-value
iMAP vs. Standard Care	**0.916**	(0.857, 0.980)	0.011

iMAP resulted in a statistically significant 8.4% reduction in the overall rate of acute health services utilization.

R software used

Conclusion

- iMAP intervention is **superior to** standard care:
  - *reduced number of MRPs at 6 and 12 months*
  - *evidence of lower risk of first health service utilization*
  - *reduced number of service utilization at 12 months*

- Study Evaluation
  - **Strength**
    - *Large sample size*
    - *Small data missingness*
  - **Weakness**
    - *Limited non-white demographic*
    - *Slightly deviated Q-Q Plot*

Acknowledgment

1. **Presenting member's order**
    a. Intro: Justin, Jianing
    b. Method: Jay
    c. SAP: Chunyi, Zhengxian
    d. Result: Huan, William
    e. Conclude: Annie

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2. **Presention slide deck building**
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    b. **Creating slides**: Justin Kim, Shujie Zhang, Chunyi Wang, Jianing Zhang, Huan Tian, William Zhang, Zhengxian Zhang, Annie Hou
    c. **Presenting**: Justin Kim, Shujie Zhang, Chunyi Wang, Jianing Zhang, Huan Tian, William Zhang, Zhengxian Zhang, Annie Hou
    d. **Answering questions**: Jianing Zhang, Annie Hou

References

1. Individualized Medication Assessment and Planning: Optimizing Medication Use in Older Adults in the Primary Care Setting
2. Development and Testing of a Tool for Assessing and Resolving Medication-Related Problems in Older Adults in an Ambulatory Care Setting: The Individualized Medication Assessment and Planning (iMAP) Tool

Thank you

Group 4